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To cite this article: Philip Furley (2021): The nature and culture of nonverbal behavior in sports: theory, methodology, and a review of the literature, International Review of Sport and Exercise Psychology, DOI: [10.1080/1750984X.2021.1894594](https://doi.org/10.1080/1750984X.2021.1894594)

To link to this article: <https://doi.org/10.1080/1750984X.2021.1894594>



Published online: 07 Mar 2021.



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REVIEW



The nature and culture of nonverbal behavior in sports: theory, methodology, and a review of the literature

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ABSTRACT

Research on body language (nonverbal behavior, NVB) in sport is only slowly emerging, although it is considered important and has been frequently studied in other disciplines. The goal of the review is to provide a bio-cultural framework, methodological guidelines, and a review of existing studies on NVB in sport. Three methodological approaches are described to investigate naturally occurring NVB: evaluative, descriptive, and automated coding. Forty five studies were identified that met the inclusion criteria (i.e. peer-reviewed empirical research using either of the methodologies to study the encoding and decoding of in-situ NVB in sports). Critical review methodology was used to integrate the existing research with the outlined bio-cultural framework. There was evidence for both biologically evolved and culturally learned influences on the encoding and decoding of NVB in sports. Results showed that athletes and referees show NVB that is correlated with various internal (e.g. emotions) or contextual (e.g. success) variables and observers can decode these NVBs. Recommendations for future research are made on how researchers can exploit the fact that NVB often informs observers about what is going on inside a person and thereby gain insights on the reciprocal relationships between contextual factors, the athlete, and performance in sports.

ARTICLE HISTORY

Received 27 July 2020

Accepted 15 February 2021

KEYWORDS

Sport; nonverbal communication; body language; emotion; facial expression

Not many television broadcasts of sports events go by without the commentator mentioning the body language of the athletes. The importance of body language (scientifically referred to as nonverbal behavior; NVB) or nonverbal communication in everyday social life is often highlighted with Paul Watzlawick's aphorism 'one cannot not communicate' (Watzlawick et al., 1967, p. 51) as people are constantly sending out nonverbal signals that convey some kind of information or message to observers and thereby influence social interactions. Although the study of NVB has been a focal topic in numerous scientific disciplines the field of sport psychology has been widely unaffected by this research and has been conducted in a piecemeal fashion (Furley & Schweizer, 2020). Therefore, the aim of this review is to introduce a theoretical framework to study and integrate existing findings from different methodological backgrounds on NVB in sports. In addition, the present review attempts to inform, structure, and motivate future research on this important facet of behavior in sports.

Defining nonverbal communication and nonverbal behavior

While nonverbal communication is often described colloquially as body language it can generally be defined as any communicative act across all modalities not expressed in words (Wiener et al., 1972). NVB is considered a subprocess of nonverbal communication (Hess, 2016; Matsumoto et al., 2013). It encompasses a wide range of behaviors as all movements can be considered expressive to some degree including facial, vocal, and postural expressions, as well as touch, proxemics, and gaze.

Theoretical framework for studying NVB in sport

One reason for the discrepancy between the frequencies in which NVB is mentioned in sports and scientific research is the lack of an overarching, integrative theoretical framework to study and understand this wide-ranging, complex facet of human behavior. The importance of an overarching theoretical framework in advancing knowledge and conducting cumulative research is well documented in science (Muthukrishna & Henrich, 2019). Although there have been attempts in the field of sport psychology to propose broad theoretical frameworks to integrate existing research findings (Davids et al., 2003; Glazier, 2017; Nitsch & Hackfort, 2010; Seifert et al., 2017; Vilar et al., 2012), these frameworks arguably have limited utility for empirically investigating NVB in sports, deriving specific predictions, and advancing cumulative understanding of NVB. For this reason, I adopted a transdisciplinary framework that integrates domain-general metatheoretical assumptions with more domain-specific middle-level theorizing (Furley, 2019; Ketelaar, 2015; Ketelaar & Ellis, 2000) to integrate research on NVB in sports with the substantial literature on NVB in other domains.

Within the present framework a theory is regarded as a middle-level explanation, lying above the level of the empirical data (that the theory attempts to explain) and below the level of higher-order assumptions (i.e. a metatheory or higher level theory) from which the theory has been derived (Ketelaar, 2015; Ketelaar & Ellis, 2000). Modern evolutionary theory has been shown to be a useful metatheoretical starting point for studying NVB and has led to an accumulation of empirical research (Al-Shawaf et al., 2016; Darwin, 1872/1988; Ekman, 2020; Shariff & Tracy, 2011). Evolutionary theory has been elaborated in detail by specifying axioms from which predictions can be derived and tested empirically (Buss, 2005; Ketelaar, 2015). Evolutionary approaches begin with the premise that observable behavior ultimately serves some function (i.e. is adaptive). In evolutionary terms the ultimate function of behavior is reproduction (the passing on of genes to the next generation). A behavior can be considered adaptive if it (and the mechanisms that produce them) increased reproductive fitness in the past, in a proximal or distal manner (Furley, 2019 for a review in sport). Obviously not every observable behavior is directly associated with reproductive fitness as many behaviors of humans are motivated by more proximate goals (e.g. Neuberg et al., 2010). In this respect, ethologists (i.e. Mayr, 1961; Tinbergen, 1972) proposed that observable behavior requires different levels of explanations: proximate and ultimate explanations. Proximate explanations describe how a behavior occurs (i.e. the mechanisms), whereas ultimate explanations describe why the phenomenon came to exist in species (i.e. functions).

Regarding the ultimate level of analysis, reproductive fitness of humans was substantially increased by group life (e.g. Brewer & Caporael, 2006; Buss, 2005; Darwin, 1872/1988;

Leakey & Lewin, 1977). Therefore, humans have evolved to become very well adapted to other humans and, as an integral part of this, have evolved brain networks (i.e. mechanisms or proximate explanations) to efficiently encode important information about their own internal states in their NVB and have evolved brain networks to accurately decode this information from others and adapt their behavior accordingly (Argyle, 1972; Darwin, 1872/1988; Ekman, 2020; Fridlund, 1994; Shariff & Tracy, 2011). Further bodily adaptations facilitate the encoding of information in NVB. For example, humans have evolved an incredible amount of facial muscles (that are no longer covered by hair to facilitate the communication of internal states; Ekman & Friesen, 1978), humans have evolved white eye-balls (facilitating detecting the direction of gaze; Tomasello et al., 2007), and have evolved the capacity to blush (presumably to communicate embarrassment to appease other group members after violating a social norm; Keltner, 2009).

A common false dichotomy in psychology is the evolution vs. learning fallacy which maps onto the ultimate-proximate distinction in describing human behavior (Al-Shawaf et al., 2019). Evolutionary explanations can be regarded as the ultimate level of analysis and learning as a proximate level. These levels of analyses are not in conflict with one another. Learning mechanisms exist because they evolved and their patterns of operation reflect the problems they evolved to solve (e.g. Barrett & Broesch, 2012). Hence, it is not surprising that both learning and culture (DePaulo, 1992) and evolved adaptations (Argyle, 1972; Darwin, 1872/1988; Ekman, 2020; Fridlund, 1994; Shariff & Tracy, 2011) have been shown to influence NVB. In this respect, the comprehensive, theoretical framework of dual inheritance theory (Boyd & Richerson, 1985; Muthukrishna & Henrich, 2019; Richerson & Boyd, 2005) is a useful theoretical specification within the metatheoretical background for the study of NVB in sport. This framework suggests that humans inherit behavior through two routes: genes and culture (dual inheritance theory: Boyd & Richerson, 1985; Richerson & Boyd, 2005). Dual inheritance theory is best described as a culture-gene co-evolutionary model that rests on the premise that human behavior is shaped by a dynamic interaction between culture and genes. As the distinction of evolved versus culturally learned influences on NVB has been central in the study of NVB, dual-inheritance theory qualifies as a suitable metatheory in comparison to other broad metatheoretical approaches that are frequently used in sport psychology (e.g. information processing theories; e.g. Neisser, 1976, embodied cognition theories; e.g. Cappuccio, 2019, or dynamical system theories; Araujo et al., 2006).

Encoding and decoding NVB

Nonverbal signal production (i.e. encoding) is not similarly rule-based as language production is, nor does it follow an explicit grammar. Nevertheless, research shows that nonverbal cues are processed highly efficiently and quickly in humans (i.e. decoding) and some NVB has been described as 'part of a universally recognized and understood code' (Burgoon et al., 1996, p. 5).

The encoding of NVB

Encoding is the act of producing nonverbal information such as facial expressions, gestures, body movements, and postures (Argyle, 1988; DePaulo & Friedman, 1998). NVB is encoded both consciously (i.e. deliberate) and unconsciously (i.e. autonomous) (Matsumoto et al., 2013). Hence, NVB is conceptualized to systematically vary along a continuum

of controllability (e.g. Ekman & Friesen, 1969; Rosenthal & DePaulo, 1979a, 1979b). For example, two distinct neural pathways have been identified for controlling facial expressions: The pyramidal tract deriving from the cortical motor strip drives voluntary facial actions, whereas the extrapyramidal tract originating in subcortical areas drives involuntary expressions (Rinn, 1984).

Following from the early work of Darwin (1872/1998), NVB is often assumed to autonomously convey information about emotion and has led to the formulation of *basic emotion theory* (Ekman & Cordaro, 2011): i.e. certain universal emotional themes (i.e. basic emotions; e.g. fear, anger, disgust, joy, and sadness) are universally expressed in a person's NVB (particularly their faces, Ekman, 2020). *Basic emotion theory* has come under increasing scrutiny since the 1990s (Fridlund, 1994; Russell et al., 2003; Russell & Fernandez-Dols, 1997) as NVB can convey many other kinds of information besides emotional information, such as values, personality dispositions, psychopathologies, physical states such as fatigue, and cognitive states such as comprehension or interest (Fridlund, 1994). The most prominent rival theory to basic emotion theory is *behavioral ecology theory* (Crivelli & Fridlund, 2018, 2019; Fridlund, 1994). According to this view, NVB is not necessarily a medium that broadcasts how an individual is feeling but should rather be considered a tool for communicating how an individual is likely to act in the future (e.g. Parkinson, 2005).

In line with dual-inheritance theory (Boyd & Richerson, 1985; Richerson & Boyd, 2005), not all NVB are autonomous, evolved expressions outside of intentional regulation. Particularly athletes are often exerting some control over their NVB when interacting with spectators, referees, team mates or opponents (DePaulo, 1992; Furley & Schweizer, 2020; Hackfort & Schlattmann, 2002). In this respect, Erving Goffman (1959, 1963, 1971) was one of the first to explicitly state that people can purposely control their NVB to convey particular impressions. In addition, a large body of research has shown that culture influences NVB in profound ways (Matsumoto et al., 2013). To account for these cultural influences on NVB, Ekman and Friesen (1969) coined the term cultural display rules that are assumed to be learned in early childhood and help individuals manage and modify their NVB depending on the social circumstances (e.g. not to express frustration but show acknowledgment of the winner during an award ceremony after losing a sport contest). However, research shows that people cannot always keep up their 'masquerade' when intentionally modifying their NVB for self-presentation purposes (DePaulo, 1992; Jones & Pittman, 1982) and leak information via autonomous NVB, especially when stressed, fatigued, or engaged in another task (Ekman, 2016; Furley & Schweizer, 2020).

The decoding of NVB

Decoding describes the process of picking up nonverbal signals and cues, making sense of these, and acting upon them (Matsumoto et al., 2013). Again, both nature and nurture seem to influence the decoding of NVB as some NVB seem to be automatically interpreted in a similar manner (Keltner et al., 2019), whereas the interpretation of other NVB is influenced by cultural learning (Argyle, 1972; DePaulo, 1992). Our cognitive system has become particularly attuned to nonverbal cues that are of general adaptive relevance (e.g. dominance or threat related information; McArthur & Baron, 1983; Zebrowitz & Collins, 1997). This means that the information processing system of humans is hard-

wired to pick up and decode this kind of information without needing any additional contribution of higher order cognitive processes (e.g. the frontal lobes of the brain). Freeman and Ambady (2011) suggest that the perception of NVB in real life involves the constant interaction among high-level categories, stereotypes, and the low-level processing of facial and bodily cues (i.e. direct perception proposed by ecological theorizing) which dynamically unfold over time.

Methodological approaches for investigating encoding and decoding NVB in sports

Three different methodologies have been used to study the decoding and encoding of NVB: Evaluative coding, descriptive coding, and automated coding (Bente et al., 2008 for further detail).

Evaluative coding

Within this approach, nonverbal expressions are shown to a diverse group of participants who are asked to evaluate the internal state of the person or make some other inference based on the NVB of the person (e.g. if the athlete is currently leading or trailing). This methodology has also been termed *thin-slices paradigm* (see Ambady & Rosenthal, 1992; Ambady et al., 2000 for an overview). The aim of this approach is to transform observed behavior into psychological categories (e.g. aggressiveness or confidence). The goal is not obtaining precise behavioral descriptions, but subjective impressions of observers (Bente et al., 2008). Relatively large number of judges are needed to achieve acceptable levels of agreement (Hess, 2016). This does not mean that this procedure generally yields invalid data, only that evaluative coding is not a very suitable means of behavior description and, consequently, as a tool for production (i.e. encoding) studies (Bente et al., 2008).

Descriptive coding

In this approach, NVB is quantified by trained coders who use a predefined descriptive coding system. The advantage of coding systems is that they give precise information on the actual bodily movements shown. However, a challenge is to assign a specific meaning to the movements. Further, this procedure is highly time consuming. The most prominent and most widely used system is undisputedly Ekman and Friesen's Facial Action Coding System (FACS, Ekman & Friesen, 1978). Methodological advancement in the field of body movement coding has been slower and proven more difficult due to the large degrees of freedom in whole body movements (Harrigan et al., 2002 for a review).

Automated coding

This approach makes use of technological devices like computer-based motion capture devices or electromyography to measure muscle activity associated with NVB (Hess, 2016). It has the benefit of being time efficient and giving excellent time and space

resolution, thereby capturing subtle expressions (e.g. facial mimicry). The downside of this newly emerging approach is that 'these devices are quite obtrusive and often impractical for use in natural settings, since they require the attachment of sensors and the installation of stationary laboratory equipment' (Bente et al., 2008, p. 272). Machine learning algorithms (e.g. Noldus FaceReader) can potentially circumvent this problem and have shown promising utility for automatically coding NVB in video clips (particularly regarding facial NVB; e.g. Lewinski et al., 2014).

Literature review of encoding and decoding NVB in sport

The context of sport has a lot to offer for research on NVB as hardly any other social context is documented and video-recorded to a similar degree and is as replete with a range of naturally occurring intense nonverbal expressions as modern sports competitions (Furley, 2019). For the present review I will exclusively focus on studies that have used either of the three outlined methodological approaches to investigate the decoding and encoding of naturally occurring NVB in the face and body within the context of sport. Another line of research in sport has investigated effects of NVB on various cognitive, emotional, behavioral or performance outcomes. I will not review studies that manipulated NVB and tested the effects of NVB on observers as this has been exhaustively reviewed in a recent handbook chapter (Furley & Schweizer, 2020).

I used critical review methodology (Grant & Booth, 2009) to summarize the respective studies and integrate them within the proposed theoretical framework. Given the ambiguity of studies on NVB in sports and the scope of behaviors qualifying as NVB (Watzlawick et al., 1967) a qualitative review approach seems more appropriate than a quantitative one as the individual studies cannot easily be reduced to comparable numbers. To ensure some comparability of study, narrow the scope of the literature review, and reduce the risk of selection bias, the following eligibility criteria for articles were included: (1) peer-reviewed publication, (2) publication in English, (3) empirical study with human participants, (4) the primary focus of the study had to be on body language, nonverbal behavior or nonverbal communication, and (5) one of the above methodological approaches was used on visually observable NVB in the face and/or body during in situ sport conditions. The rationale of focusing on the encoding and decoding of NVB in the face and body via the visual channel was that this is assumed to be the most important form of NVB in humans and has received most research attention (Hess, 2016; Matsumoto et al., 2013; Wiener et al., 1972).

Web of Science, PubMed, and EBSCOhost including (SPORTDiscus, PSYINDEX, and SocINDEX) were used as search engines. I searched the following terms combined with different expressions (nonverbal behavior OR body language OR nonverbal communication OR facial expression) AND (sport OR exercise OR physical activity). The combination of such terms was applied in each of the databases. Subsequently, I scanned the reference lists from all selected papers and further searched the reference lists of related textbook and review articles. This led to the inclusion of 11 papers with 19 studies (15 using evaluative coding, 2 descriptive coding, and 2 automated coding). Scanning the reference lists of the search results and review papers (Furley, 2019; Furley & Schweizer, 2020) led to the inclusion of 21 additional papers with 26 studies (8 using evaluative coding, 14

descriptive coding, and 4 automated coding). See [Table 1](#) for an overview of the included studies. The search was conducted in November 2020.

NVB as a consequence of prior success or failure in sports

Not surprisingly most studies on NVB in sports have chosen to investigate NVB associated with success and failure.

Evaluative coding of winning and losing NVB

In a series of studies (Furley et al., 2017, 2018; Furley & Schweizer, 2014a, 2014b; Ryan et al., 2016) different groups of participants had to estimate who was currently leading or trailing based on short video-recordings of athletes' NVB during breaks in a game. For example, participants watched brief videos from basketball matches that showed players during breaks in the game (e.g. time-outs, Furley & Schweizer, 2014a) with the score occluded. Videos were sampled according to predefined criteria by research assistants who were blind towards the hypotheses. Participants were asked to estimate whether the depicted players were currently trailing or leading on a continuous scale. The results demonstrated that observers of sport competitions (basketball, table-tennis, and European handball) could reliably decode score tendencies based on NVB without being aware of the current score.

A follow-up study showed that leading athletes were rated higher on dimensions related to social status (i.e. dominance, pride, confidence) than trailing athletes (Furley & Schweizer, 2014b). These findings can be interpreted as the result of adaptations that evolved to solve adaptive problems in our ancestral past: Sending low status signals when losing an antagonistic encounter increased the chances of avoiding further potentially life-threatening attacks. Likewise, sending victorious signals helped to save valuable resources by communicating high status and superiority over the opponent, and further signal supremacy to potential mates (Furley, 2019). Although sending submissive cues seems adaptive in primates when losing a fight because this helps avoiding further attacks, it seems unlikely that leading athletes will go easy on trailing athletes as a result of submissive NVB (Furley & Schweizer, 2014b). As athletes should be motivated to avoid sending submissive nonverbal signals during competition, this finding supports the introduced evolutionary theorizing: i.e. although athletes try to use their NVB for self-presentation purposes, their NVB can leak information that they would rather want to hide.

A recent study using event-related brain potentials (Furley et al., 2017) showed enhanced attentional orienting towards leading as compared to trailing athletes indicating that subtle signs of athletic supremacy are reliably differentiated in the human brain (i.e. decoding). This finding indicates that humans have evolved a neural receptivity towards signals associated with high status and superiority. As the desire for status can be considered a fundamental human motive, our perceptual-cognitive system seems to have become particularly attuned to status cues (see Anderson et al., 2015; Schmid Mast & Hall, 2004). This reasoning is further supported by empirical research showing that observers can distinguish between leading and trailing athletes under very challenging conditions. For example, observers can decode cues associated with leading and trailing even from briefly presented still pictures, or when either facial or bodily

Table 1. Studies included in the literature review.

Reference	Method	Participants	Main findings
NVB as a consequence of prior success or failure in sports <i>Evaluative coding of winning and losing NVB</i>			
Furley and Schweizer (2014a), Study 1	Evaluative coding of current score	40	Male and female adults (with no particular knowledge of sports) could assess who was leading or trailing based on NVB of athletes in basketball and table tennis
Furley and Schweizer (2014a), Study 2	Evaluative coding of current score	Total 40: 20 (4–8-year old) children; 20 (9–12-year old) children	Children (younger and older children were equally accurate) were able to estimate who was leading or trailing based on NVB of athletes in basketball and table tennis
Furley and Schweizer (2014a), Study 3	Evaluative coding of current score	Total 40: 20 adults with domain-specific knowledge of the sport 20 adults without domain-specific knowledge of the sport	Adults with and without domain-specific sports knowledge could both equally well assess who was trailing or leading based on athletes' NVB in the sport of handball
Furley and Schweizer (2014b), Study 1	Evaluative coding of dominance, pride, and confidence	40	Observers rated leading athletes as more dominant, more proud, and more confident than trailing athletes, without being aware of the score
Furley and Schweizer (2014b), Study 2	Evaluative coding of current score	Total 120: video-head-and-body-group ($n = 41$); video-head-only-group ($n = 39$); video-body-only-group ($n = 40$)	Perceivers were able to distinguish trailing from leading athletes based on athletes' nonverbal behavior equally well in all experimental groups, even when either athletes' heads or bodies were not visible
Furley and Schweizer (2014b), Study 3	Evaluative coding of current score	Total 120: static-head-and-body-group ($n = 41$); static-head-only-group ($n = 39$); static-only-group ($n = 40$)	Perceivers were able to distinguish trailing from leading athletes based on short presentations of 'frozen' frames depicting athletes' nonverbal behavior, even when either athletes' heads or bodies were not visible
Furley et al. (2017)	Evaluative coding of current score (+EEG)	45	Participants could reliably differentiate trailing and leading athletes based on their NVB and showed enhanced attentional orienting (EEG signal) towards leading as compared to trailing athletes
Furley et al. (2018)	Evaluative coding of current score	156	Male and female adults (with no particular knowledge of sports) could assess who was leading or trailing based on NVB of athletes in basketball and table tennis
Matsumoto and Hwang (2012), Study 2	Evaluative coding of triumph	50 US participants	Observers differentiated expressions labeled pride from triumph
Matsumoto and Hwang (2012), Study 3	Evaluative coding of triumph	62 participants from South Korea	Replication with South Korean participants. Expressions labeled as Triumph by the US participants were also labeled as Triumph by South Korean participants
Ryan et al. (2016)	Evaluative coding of current score	Total 38 children with ASD: 15 (5–8 year old) children; 23 (9–13year old) children	Children with ASD were able to distinguish trailing from leading athletes based on athletes'

(Continued)

Table 1. Continued.

Reference	Method	Participants	Main findings
			nonverbal behavior. There estimates were similarly accurate compared to typically developing children
<i>Descriptive coding of winning and losing NVB</i>			
Hwang and Matsumoto (2014)	Coding of dominance threat displays	1 Coder coded 244 images of sighted and blind Judoka (6 coders coded samples for reliability)	Winning judoka showed NVB that have previously been described as dominance threat displays: i.e. full body actions associated with body expansion, aggression, and attention
Matsumoto and Hwang (2012), Study 1	Behavioral triumph coding and evaluative coding	2 Coders coded 47 images of Judoka which were evaluatively coded by 147 participants on emotional dimensions	The combined approach of descriptive and evaluative coding procedures showed that expressions labeled triumph were distinct from other expressions: 'arm away from body', 'arm raised above shoulder', 'fist', and 'punching motion' differentiated the expressions labeled as triumph from others
Matsumoto and Willingham (2006)	FACS Coding + interpretation with Emotion FACS dictionary	2 FACS coders coded 190 photos of 85 athletes in gold and bronze medal Olympic matches	Athletes showed universal emotional expressions. Expressions differed reliably depending on the antecedents: i.e. winning vs. losing directly after the match (i.e. more autonomous regulation) or during the ceremony (i.e. more deliberate regulation)
Matsumoto and Willingham (2009)	FACS Coding	2 FACS coders coded 123 photographs of sighted and blind Judoka	No differences either in individual facial actions or in facial emotion configurations between congenitally blind, noncongenitally blind, and sighted athletes. Blind athletes showed similar emotional reactions depending on the context as sighted athletes
Matsumoto et al. (2009)	FACS Coding	2 FACS coders coded Photos of 68 Judoka	Initial emotional expressions of athletes were universal, however, their subsequent expressions were culturally regulated as these expressions were correlated with variables associated with the respective cultures of the athletes
Tracy and Matsumoto (2008)	FACS Coding + pride and shame coding system	2 FACS coders and 3 Pride/Shame coders coded 171 (111 Olympic/60 Paralympic) NVB after winning and losing Judo contests	Sighted, blind, and congenitally blind individuals from over 30 nations displayed similar pride expression in response to success and shame expressions in response to failure
<i>Descriptive coding of post-shot NVB as a function of success</i>			
Drewes et al. (2020)	NEUROGES-ELAN Coding System	2 Coders coded head movements of 17 tennis players between points	Losing players showed significantly more head movements. Losers showed more upward and sideward head movements compared to winners
Lev et al. (2020)	Ad-hoc coding scheme to measure celebratory NVB	1 Coder coded 366 situations of post-goal celebrations of professional soccer players	Most goal celebrations were interactive involving the scoring player and his teammates.

(Continued)

Table 1. Continued.

Reference	Method	Participants	Main findings
Moesch et al. (2015a)	PSB-G (Post shot behavior-gesture); PSB-T (Post shot behavior-touch)	1 Coder coded 1239 situations of post-shot behaviors of professional handball players	Interactive celebrations lasted longer than individual celebrations. Celebrations lasted longer following important goals. Celebrations differed according to the culture of the player. Development of coding scheme of post-shot NVB in handball. Reliability was acceptable. Final coding scheme: post-shot behavior gesture (PSB-G) with six behaviors and post-shot behavior touch (PSB-T) with five behaviors
Moesch et al. (2015b)	PSB-G (Post shot behavior-gesture); PSB-T (Post shot behavior-touch)	1 Coder coded 616 situations of post-shot behaviors of professional handball players	2.7 NVB were displayed after scoring a goal. One fist up, two fists, and high fives were displayed most often. NVBs were displayed more in important matches and more the higher the team was leading
Moesch et al. (2018)	PSB-G (Post shot behavior-gesture); PSB-T (Post shot behavior-touch)	1 Coder coded 616 situations of post-shot behaviors of professional handball players	The better team performance was in the preceding period, the more gestures were shown by the shooter after scoring in the following period. Nonverbal postshot celebrations were related to subsequent team performance, but only when the ongoing history of events was taken into account, and only for touch
Moll et al. (2015a)	Ad-hoc coding scheme to measure post-shot pride and shame	1 Coder coded 151 situations of post-shot behaviors of professional soccer players	Celebrations including both arms were correlated with winning the shootout. It was more likely that the subsequent penalty by an opponent was missed after a player displayed pride after a goal than when he did not
Success and failure as a consequence of NVB in sports			
<i>Evaluative coding of pre-shot NVB</i>			
Furley and Memmert (2020), Study 1	Evaluative coding of points scored	138	Observers rated the performance of Darts players higher the better the actual performance was based on pre-performance NVB
Furley and Memmert (2020), Study 2	Evaluative coding of points scored	134	Experiment 2 replicated the findings of Experiment 1 with different stimulus material. Pre-performance NVB in Darts was predictive of performance estimates
Furley and Memmert (2020), Study 3	Evaluative coding of points scored	100	Pre-performance NVB in the face of Darts players was sufficient to draw accurate inferences about performance tendencies.
Furley and Memmert (2020), Study 4	Evaluative coding of points scored	118	Pre-performance NVB of single darts hitting a triple led to overall higher performance ratings compared to ratings of the pre-performance behavior of darts not hitting a triple.
<i>Descriptive coding of pre-shot NVB</i>			
Furley and Roth (2021)	Body Action and Posture Coding System (BAP)		NVB differed for successful and missed shots in a sample of

(Continued)

Table 1. Continued.

Reference	Method	Participants	Main findings
		2 Coders coded 100 penalty preparations (69 missed and 31 scored penalties)	penalties (time spent looking towards goal, towards the ground, right arm movement, and how upright body posture was)
Jordet and Hartmann (2008)	Ad-hoc coding scheme to measure preparation NVB in penalty kicks (looking & rushing)	2 Coders coded 359 penalty kicks in soccer (25 positive valence shots; 34 negative valence shots)	Players more often turned their back to the goalkeepers and rushed through the penalty preparation more often when a shot could result in a loss of the shootout. This NVB was associated with worse penalty performance
<i>Descriptive coding of NVB prior to competition.</i>			
Kraus and Chen (2013)	FACS Coding of smile intensity	4 Coders coded smile intensity of 152 MMA fighters	Increased smile intensity before an Ultimate Fighting match was associated with poorer performance by the fighter showing the smile, and better performance by his opponent
<i>Descriptive coding of touch NVB prior and during competition</i>			
Kneidinger et al. (2001)	Touch coding	1 Coder coded 3554 touching behaviors of 119 men and 52 females	Women showed more touching behavior altogether. However, gender affected touching behavior depending on the context.
Kraus et al. (2010)	Touch coding and cooperation coding	2 Coders coded touch behavior and cooperation in 294 players of all 30 NBA teams	Touch behavior was associated with better basketball performance, even after controlling for several confounding variables. The effect of higher levels of touch on enhanced team performance was mediated by increased cooperative behavior between teammates
NVB associated with further contextual and personal variables in sport			
<i>Evaluative coding of territoriality and NVB in sports</i>			
Furley et al. (2020), Study 1	Evaluative coding of playing at home or away	80	Observers could differentiate between home playing and away playing professional soccer players based on their NVB prior to the match
Furley et al. (2020), Study 2	Evaluative coding of assertiveness, dominance, and aggression	203	Observers rated professional soccer players higher on assertiveness, dominance, and aggression when playing at home compared to playing away
Furley et al. (2020), Study 3	Evaluative coding of assertiveness, dominance, and aggression	112	Observers could differentiate if amateur players (that were not performing in front of a large crowd) were playing at home or away. Amateur soccer players were rated as more assertive, dominant, and aggressive when playing at home compared to playing away
<i>Evaluative coding of challenge and threat states and NVB in sports</i>			
Brimmell et al. (2018)	Evaluative coding of confidence, dominance, composed, focus, threatend, and shot accuracy	71	Penalty takers who evaluated a penalty as a challenge were rated as more dominant, confident, composed, challenged, and competent from their NVB than those who evaluated it as a threat

(Continued)

Table 1. Continued.

Reference	Method	Participants	Main findings
<i>Evaluative coding of NVB in sports refereeing</i>			
Furley and Schweizer (2016b), Study 1	Evaluative coding of confidence	40	Observers rated referees' NVB as less confident following ambiguous decisions as compared to unambiguous decisions
Furley and Schweizer (2016b), Study 2	Evaluative coding of confidence	60	Perceivers regarded referees' NVB as less confident following ambiguous decisions as compared to following unambiguous decisions when no context information was observable
Furley and Schweizer (2016b), Study 3	Evaluative coding of how likely one would contest decision	58	Perceivers were more likely to debate with the referee when referees nonverbally communicated less confidence following an ambiguous decision.
<i>Automated facial action coding on emotional expressions during and after sports</i>			
Hetland et al. (2018)	Noldus FaceReader	53 backcountry skiers	Participants showed less happiness during skiing compared when they were stopping for a break. Expressed emotions during an event were largely unrelated to self-report measures of emotions after the event
Hetland et al., 2019	Noldus FaceReader	24 downhill mountainbikers	Participants displayed less happiness during the activity than before or after the activity. No significant associations between facially expressed emotions and self-reported emotions were evident
<i>Automatically assessing facial NVB as a function of exercise strain and fatigue</i>			
de Morree and Marcora (2010)	EMG measurement of facial muscles	20 participants (10 fatigue group, 10 control group)	Facial EMG, RPE, and leg EMG significantly correlated with each other. Frowning muscle activity reflected effort during physical tasks
de Morree & Marcora, 2012	EMG measurement of facial muscles	20 participants (10 heavy-intensity group, 10 severe-intensity group)	Results were fairly inconclusive, but provided preliminary evidence that facial EMG could differentiate between different exercise intensities during constant-workload cycling
Huang et al., 2014	EMG measurement of facial muscles	33	Heart rate and EMG activity of the facial muscles and the primary active lower limb muscle were strongly correlated with RPE. Facial muscle activity increased significantly during high-intensity exercise
Miles et al., 2018	facial feature tracking to measure facial muscle changes	15 cyclists	Facial movement of the whole face, the upper face, lower face, and head movement increased with exercise intensity

information has been occluded (Furley & Schweizer, 2014b). Even young children (Furley & Schweizer, 2014b) and children with autism spectrum disorder (Ryan et al., 2016) were able to decode such cues when watching short video clips of athletes during competition.

Descriptive coding of winning and losing NVB

Matsumoto and Willingham (2006) examined the facial expressions of Judokas at the 2004 Olympic Games after they had won or lost a medal competition. Facial expressions were coded with the FACS and interpreted using the Emotion FACS dictionary (Ekman & Friesen, 1978). The study analyzed the facial expressions of 85 athletes. The study included photos that were taken directly after the outcome of the match was known. Photos were also included from the medal ceremonies after the matches: (a) when the athletes' received their medals and (b) when the athletes were on the gold medal stand posing. Results showed that many expressions of athletes corresponded to facial expressions that are believed to be universal. The expressions differed reliably depending on the antecedents that produced them. Winning vs. losing directly after the match (i.e. more automatic, evolved regulation) was associated with different expressions than during the ceremony (i.e. more deliberate, learned regulation).

Tracy and Matsumoto (2008) used two coding schemes to investigate spontaneous nonverbal reactions of Olympic and Paralympic athletes after winning or losing an 2004 Olympic Judo fight. The FACS (Ekman & Friesen, 1978) was used to code facial NVB. They also adapted a pride and shame coding system in which trained coders had to rate intensity of body movements (e.g. head tilted back) on five point Likert scales. These coding schemes were used on a sample of 87 athletes (42 winners, 45 losers) at Olympic Games and 53 blind competitors (30 winners, 23 losers) at Paralympic Games. Sighted, blind, and congenitally blind individuals from over 30 nations displayed similar pride expression in response to success and shame expressions in response to failure. The finding that congenitally blind individuals across cultures showed similar responses to success indicates that these NVBs have an innate component but can be intentionally modified according to cultural display rules.

Matsumoto and Willingham (2009) studied a sample of 123 photographs of sighted and blind Judoka at the 2004 Olympic and Paralympic Games for FACS coding. Results revealed that there were no differences either on the level of individual facial actions or in facial emotion configurations between congenitally blind, noncongenitally blind, and sighted athletes. Further analyses in the blind athlete sample revealed that blind athletes' expressions differentiated between whether they had won or lost a medal match at 3 different points in time, and there were no cultural differences in expression. Expressions were comparable to sighted athletes' expressions and suggest that emotional expressions serve the same functions in sighted and blind athletes and are not dependent on observational learning.

Hwang and Matsumoto (2014) again sampled images from Judo Championships (the 2004 Olympics, Paralympics, and 2008 Olympics). Images were selected by identifying the first whole body reaction occurring after the result of the match was known. Selected images were coded based on the behavioral criteria proposed to be associated with triumph expressions (Matsumoto & Hwang, 2012). Fifteen behaviors associated with triumph were coded (e.g. arms raised above shoulder, chest out, fist, head tilt back/up) and 5 behaviors assumed to be indicative of pride or shame (e.g. hands on hips, head tilted forward or down). Results were interpreted in line with evolutionary theory that the first spontaneous nonverbal reaction of winning judoka show NVB that has previously been described as dominance threat displays in other animals and can be described as full

body actions that are associated with bodily expansion, aggressive, and attentive behavior.

Matsumoto et al. (2009) FACS coded the facial expressions of 68 Judoka during the 2004 Olympics. The aim was to investigate the emotional displays across time and test the association between those expressions and a number of characteristics associated with the countries the athletes represented. Results indicated that the initial emotional expressions of athletes were universal, however, their subsequent expressions were culturally regulated as these expressions were correlated with variables associated with the respective cultures of the athletes. For example, athletes from individualistic cultures like the US were more expressive than athletes from collectivistic cultures (e.g. Japan) who were more likely to mask their emotions.

A series of studies used a combination of evaluative and descriptive coding methodologies on the images from the 2004 Olympic Judo competitions (Matsumoto & Hwang, 2012). Results showed that depending on the response options available to participants, the responses significantly differed, highlighting the importance of considering response options in designing evaluative coding studies. Further, behavioral coding indicated that the expressions evaluated as triumph by participants were distinct from other expressions. Results showed that 'arm away from body', 'arm raised above shoulder', 'fist', and 'punching motion' differentiated the expressions labeled as triumph from others. Results from two additional studies suggested that athletes from different cultures seem to encode triumph nonverbally in a similar manner and observers from different cultural backgrounds universally decode this as triumphant behavior. The authors interpreted these findings as evidence for (a least partly) evolved spontaneous expressions of triumph.

Descriptive coding of post-shot NVB as a function of success

A series of studies with an applied focus developed the Handball Post-Shot Behavior Coding Scheme (H-PSB-CS; Moesch et al., 2015a) to code post-shot behaviors by female handball players. Moesch et al. (2015b) used the developed coding system to explore contingencies between post-shot NVBs in handball and events during the game. Six hundred and sixteen coding situations were analyzed that had resulted in a goal. On average about 2.7 NVBs were displayed by players after scoring a goal (about 1.5 NVB involving gestures and 1.2 NVB involving touch between team-mates). One fist up and two fists up were the gestures displayed most often. The touch behaviors exhibited most often after scoring were high fives. NVBs were displayed more often in play-off matches than league games, more NVBs were displayed after a goal the higher a team was leading, and the overall amount of NVBs declined over the course of a match. Moesch et al. (2018) further tested if prior performance predicted the display of nonverbal post-shot celebrations and if nonverbal postshot celebrations predict subsequent performance. Results showed that the better team performance was in the preceding period, the more gestures were shown by the shooter after scoring in the following period. Nonverbal postshot celebrations were related to subsequent team performance, but only when the ongoing history of events was taken into account, and only for touch. In general, the obtained effects were very small (explaining less than one percent of variance in performance) and need to be corroborated by further research.

Moll et al. (2015a) developed an ad-hoc coding scheme to code post-shot NVBs following soccer penalty kicks in shoot-outs. The study included the post-shot behavior of 151 post-shot behaviors during penalty shootouts in the World- and European Cup. Only shots were included where the score in the shootout was tied. Results indicated that certain celebratory post-shot behaviors were more likely to be shown in the team that ultimately won the penalty shootout. Celebrations including both arms were correlated with winning the shootout. In addition, it was more likely that the subsequent penalty by an opponent was missed after a player displayed certain celebratory behaviors after a goal than when he did not.

A recent study (Lev et al., 2020) developed an ad-hoc coding scheme to investigate the celebratory NVB ($N = 366$) among professional football players in the 2018/2019 European Champions League after scoring a goal. Results revealed that most celebrations were performed interactively with teammates. When celebrations were interactive they lasted longer than when the NVB was shown solitary. The celebrations also lasted longer when following an important goal. There were some interesting cultural differences in celebratory NVB amongst players (e.g. African players demonstrated more NVB associated with religious symbols than European players).

Drewes et al. (2020) used a nonverbal coding system that was developed to measure NVB in clinical settings (NEUROGES-ELAN, Lausberg & Sloetjes, 2016) to identify specific NVBs associated with winning and losing points in tennis. Spontaneous nonverbal head movements of professional tennis athletes between points were video recorded and coded by two certified raters. Results showed significant NVB differences as a function of winning or losing points in tennis. Players losing points showed more head movements and more upward and sideward head movements than winners.

Although the studies reviewed in this section did not refer to the introduced bio-cultural theorizing, they point to interesting correlations of NVB and performance, highlighting the importance of NVB in guiding social interactions. The next section summarizes research that has investigated associations between prior NVB and subsequent sports performance, either via a direct individual pathway (i.e. internal states that might be associated with performance tendencies show in an athletes NVB) or via an indirect social pathway (NVB prior to performance influences team-mates or opponents and thereby influences performance).

Success and failure as a consequence of NVB in sports

Research has utilized evaluative and descriptive coding methodology to investigate correlations between NVB prior to sports performance and performance across several sport contexts.

Evaluative coding of pre-shot NVB

Furley and Memmert (2020) addressed the question if it is possible to predict performance tendencies based on athletes' NVB directly before the performance by randomly sampling video clips from all throws by professional Darts players in the 2017 World Championships. These short video recordings of players immediately before throwing darts were shown to participants who had to estimate the points scored for the respective throws (i.e. a series of three darts in immediate succession). In a first study, 20 video clips in

four performance categories (poor performance; medium-to-good performance; good performance; and perfect performance) were shown to participants who rated the scores on a continuous scale ranging from 0 to 180 (the scoring range in darts). The results showed that pre-performance NVB in Darts informed observers' ratings of performance tendencies of professional Darts players. There was a linear trend in performance estimates as a function of performance category as score estimates were higher the better the performance category was. A second study replicated the first study with a different set of stimulus material and provided further support for the hypothesis that pre-performance NVB in Darts was predictive of performance estimates. A third study replicated the findings with only facial information displayed. A fourth study showed that the pre-performance NVB of single darts hitting a triple led to overall higher performance ratings compared to ratings of the pre-performance behavior of darts not hitting a triple. These findings are indicative that the NVB of professional Darts players inform observers about subsequent performance tendencies.

Descriptive coding of pre-shot NVB

Jordet and Hartmann (2008) developed an ad-hoc coding scheme to quantify aspects of the penalty preparation of soccer players. The coding scheme dichotomously coded looking behavior as either approach looking (facing the goalkeeper when walking back for the run-up after placing the ball on the penalty mark) and avoidance looking (turning around from the goalkeeper while walking back). Second, preparation speed was measured as the time from the referees' signal that the shot could take place until the shooter began his run-up. Both these variables were assessed by two independent coders in 395 penalty kicks from penalty shootouts. The results showed that avoidance NVB occurred more during negative valence shots (shots that could lead to a loss) than with positive shots (shots that could lead to a win). Players more often turned their back to the goalkeepers and rushed through the penalty preparation more often when a shot could result in a loss of the shootout. Both of these NVB were associated with worse penalty performance.

Furley and Roth (2021) adapted the Body Action and Posture Coding System (BAP, Dael et al., 2012) to the context of soccer penalties, validated it, and initially used this system (NBCSP) to explore fine-grained NVB in penalties. A first validation study demonstrated the coding system could differentiate certain postures and behaviors as a function of emotional valence (i.e. positive vs. negative emotional states). Study 2 identified differences in NVB for successful and missed shots in a sample of penalties (time spent looking towards goal, towards the ground, right arm movement, and how upright body posture was).

Taken together, the studies reviewed in this section indicated that athletes show distinguishable NVBs prior to successful or not successful performance, presumably because these NVBs coincide with internal states that are facilitative or debilitating to sports performance and/or by influencing the performance of the opponent (in this case the goalkeeper in the penalty studies; Furley et al., 2012).

Descriptive coding of NVB prior to competition

FACS was used to code smiles of Ultimate Fighting athletes before competitions (Kraus & Chen, 2013). Smile intensity was exclusively coded on the zygomatic major muscle to

classify smiles into three categories: a neutral expressionless face showing no contraction of the zygomatic major, a toothless smile with moderate contraction of the zygomatic major, and a teeth-baring smile showing intense contraction of the zygomatic major. Subsequently the study tested whether smiles before a physical confrontation (Ultimate Fighting Championships; UFC) are a nonverbal signal of reduced hostility, physical dominance, and aggression. The study coded smiles of 152 fighters one day prior to the contest. Results showed that increased smile intensity before an Ultimate Fighting match was associated with poorer performance by the fighter showing the smile and better performance of the opponent.

Descriptive coding of touch NVB prior and during competition

Kraus et al. (2010) developed a procedure to code physical touch in basketball (i.e. fist bumps, high fives, chest bumps, leaping shoulder bumps, chest punches, head slaps, head grabs, low fives, high tens, full hugs, half hugs, and team huddles). Touch behavior was coded by two coders of 294 players of all 30 NBA teams. Further, six coders coded the amount of cooperation with an established coding procedure (e.g. Ellyson & Dovidio, 1985) exhibited by each of the 30 NBA teams using the same games as those used for touch codes. The study showed that touch behavior was associated with better basketball performance. The effect of higher levels of touch on enhanced team performance was mediated by increased cooperative behavior between teammates. A further study investigated gender differences in touching behavior in sports (Kneidinger et al., 2001). All on-field touches between team members were recorded and coded with a coding scheme designed for this study. Results revealed that women showed more touching behavior altogether while revealing further interesting gender difference depending on specific sport contexts.

NVB associated with further contextual and personal variables in sport

Another line of research on NVB in sport has begun to investigate contingencies between NVB and certain contextual and personal variables that are not directly related to performance. Although these studies were motivated by different theoretical backgrounds and studied a wide range of NVBs in diverse situations, they all tested the central prediction of the bio-cultural framework that 'what we see outside tells us a lot about what's happening inside an individual's head and heart' (Beckoff, 2007, p. 1; see also Ambady et al., 2000; Ambady & Rosenthal, 1992 for reviews).

Evaluative coding of territoriality and NVB in sports

Furley et al. (2020) tested if athletes would show different NVB before a competitive match depending on whether they were playing at home or away. In study 1, participants watched short videos depicting soccer players prior to a UEFA Champions League match and rated whether athletes were more likely to be playing at home or away based on their NVB. Both male and female participants could significantly differentiate between home playing and away playing athletes based on their NVB prior to the match. Study 2 showed that perceivers rated professional soccer players higher on assertiveness, dominance, and aggression when playing at home compared to playing away. Study 3

replicated the results in amateur soccer players that were not performing in front of a large crowd.

Evaluative coding of challenge and threat states and NVB in sports

Brimmell et al. (2018) tested if observers of a penalty kick preparation could detect challenge and threat states in the soccer player. Results showed that penalty takers who evaluated the task as a challenge (coping resources exceed task demands) were perceived as more dominant, confident, composed, challenged, and competent from their NVB than those who evaluated it as a threat (task demands exceed coping resources). However, physiological measures of the penalty takers were not correlated with observers' ratings of NVB. Thus, the findings of this study only partly supported the hypothesis that internal states of a person show in a person's NVB.

Evaluative coding of NVB in sports refereeing

Furley and Schweizer (2016b) investigated whether soccer referees' NVB when communicating a decision (e.g. giving a yellow card) differed based on the ambiguity of their decisions and whether perceivers could detect these differences in NVB. Videos from professional referees were extracted in a two-step process to obtain recordings of referees' NVB when communicating either an unambiguous decision (this was verified by a sample of referees who all agreed on what was the right decision in the preceding situation would be) or an ambiguous decision (sample of referees gave diverging responses about the correct decision). Videos were cut to only show the NVB of the referee communicating the decision. The results provided evidence that perceivers regarded referees' NVB as less confident following ambiguous decisions as compared to following unambiguous decisions, even though they only got to see the referee's NVB while communicating the decision and not the situation leading up to the decision. A follow-up study replicated the first study with digitally altered stimulus material to ensure that only the referees' NVB was visible and no other context information (e.g. players). A final study suggested that perceivers were more likely to debate with the referee when referees non-verbally communicated less confidence following an ambiguous decision.

Automated facial action coding on emotional expressions during and after sports

A study by Hetland et al. (2018) used FaceReader (Noldus Information Technology, Wageningen, The Netherlands) to measure facially expressed emotions during a backcountry skiing event. This technology has the advantage of saving a vast amount of time compared to descriptive coding procedures and, thus allows to analyze long video clips of facial NVB. Fifty three backcountry skiers were filmed with a head mounted camera during a skiing event. One of the main findings of the study was that participants showed less happiness during skiing than when they were stopping for a break. Further the study found that expressed emotions during an event were largely unrelated to self-report measures of emotions after the event. A further study by the same lead author used a similar approach to measure facially expressed emotions during a downhill mountain bike descent (Hetland et al., 2019). The results revealed remarkable similarities with the earlier study in skiing as participants displayed less happiness during the activity than before or after the activity. Again, no significant associations between facially expressed emotions and self-reported emotions were evident. In this

respect, the studies are not in line with the prediction that NVB informs about what is going on inside an individual. However, caution is warranted regarding these findings as NVB was measured for long durations during the sporting events and subsequently averaged and compared with one retrospective report of how the athlete remembered to be feeling. Validation studies outside of sports have shown promising results regarding the validity of the FaceReader software for emotion identification (Lewinski et al., 2014).

Automatically assessing facial NVB as a function of exercise strain and fatigue

Miles et al. (2018) used facial feature tracking to measure facial muscle changes as a function of exercise intensity. Fifteen cyclists completed three, incremental intensity, cycling trials to exhaustion while their faces were recorded with video cameras. The results showed that facial movement of the whole face, the upper face, lower face, and head movement increased with exercise intensity. The authors concluded that the results demonstrate the potential utility of facial feature tracking as a non-invasive, psychophysiological measure to potentially assess exercise intensity based on facial NVB.

de Morree and Marcora (2010) measured EMG activity of the frowning muscle (bilateral corrugator supercilii) while performing leg extensions at four different workloads. Two groups of participants were recruited for the study: the fatigue group repeated leg extensions after fatiguing exercise, and the control group repeated just the leg extensions. Facial EMG amplitude, ratings of perceived effort (RPE), and leg EMG amplitude were measured. Facial EMG, RPE, and leg EMG all correlated significantly. These findings were interpreted that specific NVB (i.e. the activity of the frowning muscle) reflects effort during physical tasks. A follow-up study using a similar protocol induced effort via constant-workload cycling instead of leg extensions (de Morree & Marcora, 2012) while measuring facial EMG of the corrugator supercilii muscles, rating of perceived effort, heart rate, and blood lactate concentration. Twenty recreationally active male and female volunteers performed a constant-workload time to exhaustion task on a cycle ergometer. Ten participants were randomly allocated to the heavy-intensity or the severe-intensity group. The results showed that facial NVB could differentiate between two exercise intensities during constant-workload cycling, although some of the findings were inconclusive.

Huang et al. (2014) measured facial EMG of the corrugator supercilii (frowning), masseter and vastus lateralis (jaw clenching) muscles during cycling exercise. Participants performed continuous incremental workload cycling exercise until they were exhausted. The results revealed that heart rate and EMG activity of the facial muscles and the primary active lower limb muscle were strongly correlated with RPE. The findings again suggested that facial NVB increased significantly during high-intensity exercise.

General discussion

The aim of the review was to introduce a theoretical framework and available methodology for investigating and explaining NVB in sport. In summary, the reviewed studies provide converging evidence that athletes and referees often display (i.e. encode) NVB that is correlated with that person's internal states and contextual variables. Observers can decode these NVBs and draw meaningful inferences from these NVB.

In line with the introduced bio-cultural framework, research showed that observers could tell if athletes were leading or trailing based on their NVB (Furley & Schweizer, 2014a, 2014b, 2014b, 2016b; Furley et al., 2018; Ryan et al., 2016) and that leading athletes were rated as more dominant, confident, and proud in comparison to trailing athletes. Similarly, athletes who won a competition showed triumph in their NVB and this was reliably evaluated by observers (Matsumoto & Hwang, 2012). An extensive series of studies of the NVBs after winning or losing Olympic or Paralympic Judo matches (Hwang & Matsumoto, 2014; Matsumoto et al., 2009; Matsumoto & Willingham, 2006, 2009; Tracy & Matsumoto, 2008) provided evidence for a specific (universal) pattern of NVB in the face and body that both blind and sighted athletes show after winning and losing. However, these (potentially universal) expressions directly after competition can be deliberately modified according to display rules (e.g. during ceremonies) and differed depending on the culture of the athletes (Matsumoto et al., 2009). These studies suggest that humans are biologically prepared in advance of experience (Marcus, 2014) to encode and decode certain information in their NVB. However, both the encoding and decoding of NVB can be modified by learning and cultural experience. A limitation of these studies was that they were all conducted by only two groups of researchers and, hence, would benefit from replications from independent labs.

Further studies provided evidence that NVB either directly prior (Furley & Memmert, 2020; Furley & Roth, 2021; Jordet & Hartmann, 2008) or one day prior (Kraus & Chen, 2013) to performance in darts, soccer or Ultimate Fighting was correlated with performance in the respective sports. Such evidence might be interpreted as supportive of *behavioral ecology theory* (Crivelli & Fridlund, 2018, 2019; Fridlund, 1994) that NVB evolved as a means for informing others about how an individual is likely to act in the future. The reviewed findings that soccer referees show confidence in their NVB depending on whether they are communicating an ambiguous or an unambiguous decision (Furley & Schweizer, 2016b) or that soccer players show challenge and threat states in their NVB (Brimmell et al., 2018) might also be supportive of the basic tenet of *behavioral ecology theory* that NVB serves as a proxy to observers as to how an individual is likely to behave. In this respect, being able to encode and decode exhaustion or fatigue as was evident in several studies (de Morree & Marcora, 2010, 2012; Huang et al., 2014; Miles et al., 2018) can arguably also be considered adaptive from an evolutionary perspective. Further research showed that specific touching NVB was beneficial in increasing cooperation and performance in team sports (Kraus et al., 2010). These studies can be considered supportive of evolutionary theorizing suggesting that touching behavior evolved in social animals to support group cohesion and cooperation, which is important in sports teams (reviewed in Furley, 2019). The finding that home playing athletes were perceived as more dominant, confident, and assertive prior to a game compared to away playing athletes (Furley et al., 2020) might also be indicative of a territoriality response that evolved in species to signal intruders to stay away from their perceived territory.

Studies with an applied focus were able to identify specific NVBs that were correlated with several context variables in sports (Moesch et al., 2015a) while further indicating interesting contingencies between these NVBs and performance in various sport contexts (Drewes et al., 2020; Moesch et al., 2015b, 2018; Moll et al., 2015a). Although these studies did not mention evolutionary theory in interpreting their findings a recent review paper in sports (Morgulev & Avugos, 2020) argued that contingencies between behavior and NVB

might be a means of how ‘success breeds success’ in nature and in sports. More specifically, Morgulev and Avugos (2020) concluded that in human competitions, just like in nature, winners experience similar responses to success, and signal their recent victory via NVB to observers. In this respect, NVB could prove to be an important mosaic piece in explaining upward or downward spirals of Matthew effects (Merton, 1968; the rich get richer and the poor get poorer), Pygmalion effects (Rosenthal & Jacobson, 1968; self-fulfilling prophecies), momentum (Iso-Ahola & Dotson, 2016; Morgulev & Avugos, 2020) or performance dynamics (Vilar et al., 2012) in sport. Future research on NVB in sports might want to scrutinize the reciprocal relationship of NVB and sport performance by looking into different pathways of how NVB and performance interact: e.g. effects on athletes, teammates, opponents, referees, or spectators.

The three outlined methodological approaches (evaluative, descriptive, and automated coding) are useful tools for advancing both basic and applied research on NVB in sports. Each approach has notable strengths and weaknesses (Bente et al., 2008 for more detail) and is better suited to answer different research question. Evaluative coding is more suited for confirmatory research on the decoding of NVB (e.g. what impressions do certain NVBs create in other people), whereas descriptive and automated coding procedures are better suited for exploratory encoding research (e.g. what specific NVBs are typically shown in certain circumstances). Evaluative Coding has the advantage of enhancing understanding on the shared meanings of certain NVBs but require relatively large numbers of research participants. Descriptive coding is proven to be very effective in reliably capturing precise NVBs (e.g. how are certain inner states encoded). However, this methodology is relatively costly and time-consuming and only short excerpts of NVB can be analyzed. In this respect automated coding procedures have advantages, although current technological devices can only capture a limited range of NVBs and questions of reliability and validity need further scrutinizing.

In conclusion, exploiting the fact that what we see on the outside tells us a lot about what is going on inside an athlete seems to offer plenty of future research opportunity on important basic and applied research questions in the field of sport psychology (e.g. on the link between emotions and behavior or interpersonal coordination). It is my hope that the introduced theoretical framework and methodological guidelines will lead to increased research on NVB and nonverbal communication in sport.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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